

Khondakar Ashik Shahriar

Machine Learning Engineer | Research Assistant

Bangladesh University of Engineering and Technology (BUET)

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Google Scholar | ResearchGate | ORCID | Github | Kaggle

Education

Bangladesh University of Engineering and Technology (BUET), Bangladesh

BSc. in Electrical and Electronic Engineering

January-22, 2022 – June-15, 2026

Thesis Supervisor: Dr. Maruf Ahmed

CGPA: 3.61/4.0

Research Experience

University of Illinois Chicago (CMRR Lab)

August 2024 - Present

Remote Research Assistant

PI: Dr. Enamul Hoque Bhuiyan

Publications

Journal Articles

1. **Glioma Grade Classification: Hypertuned CNN with Saliency-Based Insight from MRI.** *IEEE Access*, 2025.
[Khondakar Ashik Shahriar](#), Enamul H Bhuiyan*, Md Forhad Hossain, Rakibur Rahman, Muhammad EH Chowdhury
2. **Lightweight Domain-Adaptive Modulation Classification Under SNR and Channel Mismatch,** *IEEE Wireless Communication Letters*, 2026.
[Khondakar Ashik Shahriar*](#)
3. **Why Nonlinear Models Matter: Unified Analysis of Cognitive Load, Stress, and Exercise Using Wearable Physiological Signals,** *Physiological Measurements IoPScience*, 2026.
[Khondakar Ashik Shahriar*](#)
4. **Generalize Beyond the Lens: Deep Magnification Adaptation in Breast Cancer Histopathology,** *IJACIS, Springer*, 2026.
[Khondakar Ashik Shahriar](#), Maruf Ahmed, Sakib A Hossain, Muhammad EH Chowdhury, Enamul H Bhuiyan*
5. **Lightweight Privacy-Preserving Human Activity Recognition from CSI Data using CNN-Temporal Attention Network,** *Digital Discovery, Royal Society of Chemistry*, 2026.
[Khondakar Ashik Shahriar](#), Maruf Ahmed, Hafiz Imtiaz*
6. **Supervised autoencoder for gradient and BCG artifact removal in EEG during simultaneous EEG-fMRI,** *Magnetic Resonance Imaging, Elsevier*, 2026.
[Khondakar Ashik Shahriar](#), EH Bhuiyan*, Qingfei Luo, MEH Chowdhury
7. **Uncertainty-aware personalized estimation of Parkinson's disease severity from longitudinal speech,** *PLOS One*, 2026.
[Khondakar Ashik Shahriar*](#)
8. **Cross-Spectral Second-Order Modeling for Robust Event Detection in Distributed Fiber Channels,** *IEEE Communications Letters*, 2026.
[Khondakar Ashik Shahriar*](#)
9. **A physics-constrained Bayesian framework for QSM with uncertainty quantification and variational susceptibility source separation,** *Physics in Medicine & Biology*, 2026.
[Khondakar Ashik Shahriar](#), Maruf Ahmed*

Conference Proceedings

1. **Bidirectional EEG–fMRI Reconstruction via a Shared Latent Space for Cross-Modal Neuroimaging (ISMRM-2026).**
Khondakar Ashik Shahriar, Maruf Ahmed, Enamul H. Bhuiyan*
2. **Multimodal Physiological Signal Classification from Wearables: Towards Interpretable Stress and Exercise Recognition (IEEE SPICSCON- 2025).**
Khondakar Ashik Shahriar*

Research Interests

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- | | |
|---------------------------------------|--------------------------------|
| • Machine Learning | • Biomedical Signal Processing |
| • Deep Learning | • Medical Imaging |
| • ML for Wireless Communications | • Computer Vision |
| • Privacy-preserving Machine Learning | • Edge AI |

Professional Experience

ACI PLC

Position: Machine Learning Engineer

August 2026 – Present

Journal Reviewer

- IEEE Transactions on Vehicular Technology
- Measurement Science and Technology (IOP Publishing)
- PLOS ONE
- Medical Engineering & Physics (Elsevier)

Leadership

- **Vice President**, IEEE Student Branch, BUET
- **Vice Chairperson**, IEEE Robotics and Automation Society (RAS) Student Branch, BUET
- **Vice President (Technical)**, BUET Robotics Society

Technical Skills

Programming C, C++, Embedded C, Python, Verilog, MATLAB, MicroPython, Arduino, HTML, JavaScript, CSS

Research: LATEX, Microsoft Word, R, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Corresponding Authorship

Design: System Integration, Data Acquisition, React, Node.JS, FastAPI, Version Control; (Git/Github), PCB Design (Proteus, EasyEDA)

Device: Arduino Uno, Arduino Mega, Arduino Nano, ESP-32, ESP-CAM, Raspberry PI Pico, Raspberry PI, FPGA Board, STM-32.

Honors and Awards

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- IEEE VIP CUP 2025 - (2nd Runner-Up-Team EyeQ).
 - KUET BitFest Datathon - Champion.
 - EEE Day BUET 2023 Project Showcasing - 2nd Prize.
 - Undergraduate Dean's List Award (level-1/Term-1).
 - 5th in National Physics Olympiad, Bangladesh(2018).

English Proficiency

TOEFL iBT: 4.5/6.0 [[pdf](#)]

(R:4.5, L:4.5, S:4.5, W:5)

References

Dr. Enamul Hoque Bhuiyan

Research Scientist, Department of Radiology, University of Illinois Chicago

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Research Supervisor

Dr. Hafiz Imtiaz

Professor, Dept. of EEE, BUET

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Research Collaborator

Dr. Maruf Ahmed

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Thesis Supervisor

Dr. Ziaur Rahman Khan

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Undergraduate Advisor